INCH-POUND

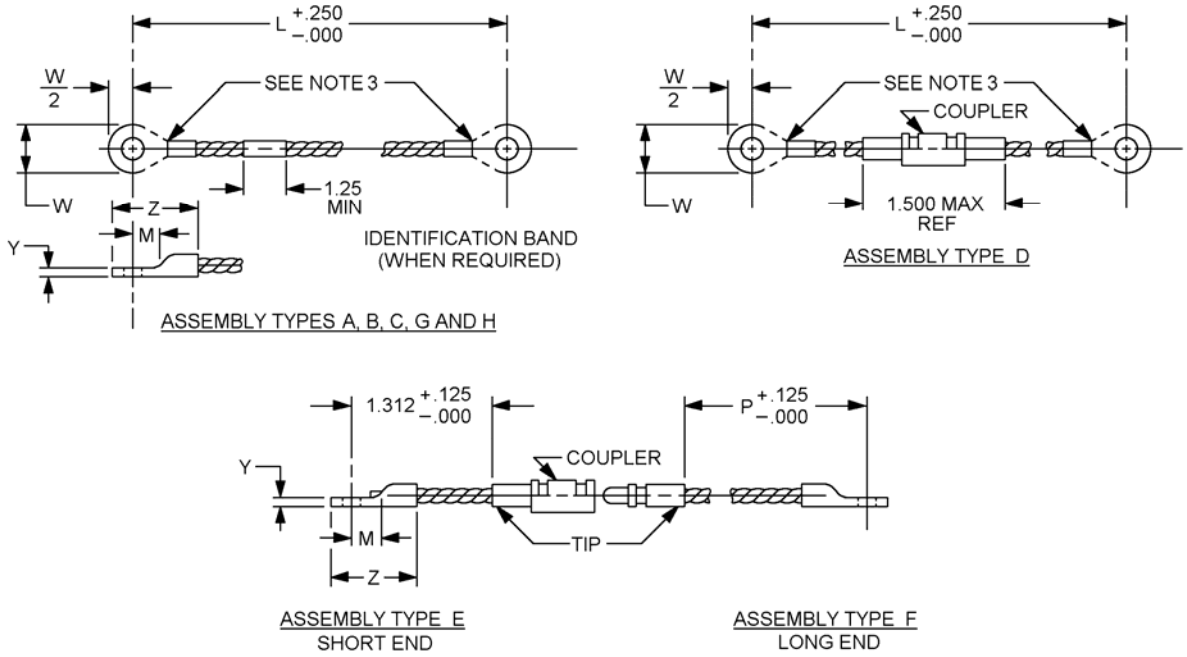
MIL-DTL 83413/8C
24 September 2008
SUPERSEDING
MIL-DTL-83413/8B
1 June 2006

DETAIL SPECIFICATION SHEET

CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: TYPE IV JUMPER CABLE ASSEMBLY, LEAD, ELECTRICAL

This specification is approved for use by
all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.



Inches	mm
.125	3.18
.250	6.35
1.25	31.75
1.312	33.32
1.500	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Resistance readings to be taken at the junction of terminal barrels and tongues.

FIGURE 1. Jumper cable assembly.

AMSC N/A FSC 6150

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TABLE I. Jumper assembly types and characteristics.

Jumper assembly type	Construc-tion type	Wire and terminal material	Wire size (see table II)	Available lug sizes (see tables III thru V)	Available cable length, see "L" or "P" dimensions on figure 1	Lightening tested	Fuel Compat-Ible 1/	Method of attaching terminals	Color of ID band	Supersedes
A	Bonding	Copper	12	A thru J	002-999	No	No	Crimp	No band	MS25083-2
B	Current return	Copper	8	K,L,M,N	002-999	No	No	Crimp	No band	MS25083-4
C 2/	Bonding	Aluminum	10	O,P,Q,R,S (aluminum only)	003-999	No	Yes	Brazed 4/	Clear	MS25083-1 and -7
D 3/	Quick disconnect	Copper	12	A, B, C,D,E	003-999	No	No	Crimp	No band	MS25083-3
E 3/	Short end of quick disconnect	Copper	12	A, B, C, D,E	Fixed length	No	No	Crimp	No band	MS25083-3S
F 3/	Long end of quick disconnect	Copper	12	A, B, C, D,E	002-999	No	No	Crimp	No band	MS25083-3P
G	Bonding	Copper	12	A thru J	003-999	Yes	No	Crimp	Yellow	MS25083-5
H	Current return	Copper	8	K,L,M,N	003-999	Yes	No	Crimp	Yellow	MS25083-6

1/ Only fuel compatible jumper assemblies may be used within fuel tank and cells.
2/ Do not use outside fuel tanks and cells.
3/ Inactive for Air Force Airborne use after 9 December 1963.
4/ Dip braze terminals in accordance with AWS C3.7 Specification for Aluminum Brazing. A coating is not required on interior strands of the cable, chemical conversion coating In accordance with MIL-DTL-5541, class 3 shall be applied after brazing. Inert gas welding using an aluminum based solder or other means of termination may be used only if approved by the qualifying activity. Aluminum terminals may be crimped to the cable to facilitate dip brazing. Over crimping shall be avoided.

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TABLE II. Wire characteristics by jumper assembly type.

Jumper assembly type	Material	Stranding no. x AWG	Wire size	Assembly 1/ resistance (max-ohms)	Tensile strength (min-lb)	Copper only	
						Crimping tools 2/	Crimping dies
A, G	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	110	M22520/5-01 3/ or M22520/24-01	M22520/5-100
B, H	Copper	(7 x 95) x 36	8	.0006xL + .00016 Ω	225	MS25441	MS23002 3/ MS90485
C	Aluminum	37 x 0.0167 inch	10	.00042xL + .00110 Ω	75	Not applicable	Not applicable
D	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	110	M22520/5-01 M22520/24-01	M22520/5-100
E	Copper	(7 x 37) x 36	12	.00057	110	M22520/5-01 M22520/24-01	M22520/5-100
F	Copper	(7 x 37) x 36	12	.00016xL + .00034 Ω	110	M22520/5-01 M22520/24-01	M22520/5-100

1/ Where L = length of the cable assembly in inches. Example: For a 24 inch type A assembly, assembly resistance = .00016 ohm/inch x 24 inches + .00034Ω = .00418 ohms (max).

2/ Alternate crimping tools which use the specified dies of this table or qualified class II copper terminal lugs in accordance with SAE-AS7928 (crimped with the tools and dies used to obtain the class II terminal lug approval), may be used to facilitate high volume production of copper jumpers. However, either of these options must be pre-approved by the qualifying activity before qualification tests are initiated on samples produced using these termination options.

3/ Application: M22520/5-01, /5-100 and MS23002 for use on insulated terminals. M22520/24-01 and MS90485 for use on uninsulated terminals.

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TABLE III. Available lug sizes.
A ,D,E,F & G type jumper assembly. 1/ 2/ 3/ 4/

Code letter lug size designation	For stud size	Copper terminals (All types except type C)	
		SAE-AS25036 Insulated	SAE-AS20659 Uninsulated
A	.138 (3.51)	-111	…
B	.138 (3.51)	…	-165
C	.164 (4.17)	-156	…
D	.190 (4.83)	-112	…
E	.190 (4.83)	…	-105
F	.250 (6.35)	-157	…
G	.312 (7.94)	-113	…
H	.312 (7.94)	…	-106
I	.375 (9.53)	5/-114	…
J	.375 (9.53)	…	-128

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ± .005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.
4/ Lightning current tests applicable to type G and H assemblies only.
5/ Size #10 & #12 AWG wire only.

TABLE IV. Available lug sizes.
B & H type jumper assembly. 1/ 2/ 3/ 4/

Code letter lug size designation	For stud size	Copper terminals (All types except type C)	
		SAE-AS25036 Insulated	SAE-AS20659 Uninsulated
K	.312 (7.94)	5/ -117	…
L	.312 (7.94)	…	5/ -108
M	.375 (9.53)	-118	…
N	.375 (9.53)	…	5/ -129

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ± .005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.
4/ Lightning current tests applicable to type G and H assemblies only.
5/ Size #8 AWG wire only.

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TABLE V. Available lug sizes.
C type jumper assembly. 1/ 2/ 3/

Code letter lug size designation	For stud size	Aluminum terminal				
		Stud Hole ID	M min	W	Y	Z max
O	.138 (3.51)	.152 (3.86)	.218 (5.54)	.406 (10.31)	.083 (2.11)	1.172 (29.77)
		.142 (3.61)		.313 (7.95)	.037 (.939)	
P	.190 (4.83)	.203 (5.16)	.250 (6.35)			
		.193 (4.90)				
Q	.25 (6.35)	.285 (7.24)	.281 (7.14)	.540 (13.72)		1.300 (33.02)
		.260 (6.60)		.450 (11.43)		
R	.312 (7.94)	.343 (8.71)	.329 (8.36)	.741 (18.82)	.090 (2.29)	1.390 (35.31)
		.320 (8.13)		.531 (13.49)	.038 (.965)	
S	.375 (9.53)	.385 (9.78)	.343 (8.71)			1.626 (41.30)
		.410 (10.41)				

1/ Dimensions are in inches. Metric equivalents are given for information only.
2/ Unless otherwise specified, tolerance is ± .005.
3/ Resistance readings to be taken at the junction of terminal barrels and tongues.

REQUIREMENTS:

Material:

Copper wire (except type C assembly): In accordance with A-A-59551, type R, class undesignated (36 AWG strands), temper 1, coating T of the size specified in table II.

Aluminum wire (type C assembly): Size 10, 37 strands of 0.0167 inch, electrical grade, aluminum alloy, 1350 strands in accordance with ASTM B230/B230M, concentric stranded, length of lay (pitch) .75 to 1.25 inch.

Copper terminals (except type C assembly): In accordance with SAE-AS7928, under the applicable SAE-AS sheet as specified in table III through table V.

Aluminum terminals (type C assembly): Aluminum alloy in accordance with SAE-AMS-QQ-A-225, ASTM-B211 or SAE-AMS-QQ-A-250 and ASTM-B209, temper O with dimensions as specified in table V.

Coupler (assembly types “D”, “E” and “F” only): In accordance with SAE-AS6852.

Identification band (types C, G and H only): Polyolefin, heat shrinkable in accordance with SAE-AMS-DTL-23053/5. Color specified in table I.

Assembly resistance: See table II.

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Tensile strength: See table II. Perform after the flexibility test in accordance with MIL-DTL-83413, group II qualification sequence. Omit assembly resistance as a tensile strength post test.

Crimping tools and crimping dies: See table II.

Marking (types C, G and H only): Part or Identifying Number (PIN) shall be marked on the identification band. If the identification band is too short to accommodate the entire PIN on a single line, the PIN may be put on two lines or both sides of the identification band. Assembly type “C” parts shall be marked with an ink or fluid which is compatible with fuel.

Salt spray: 48 hours in accordance with test procedure EIA-364-26.

Altitude-low temperature: Not applicable.

Dust: Not applicable.

Ozone exposure: Not applicable.

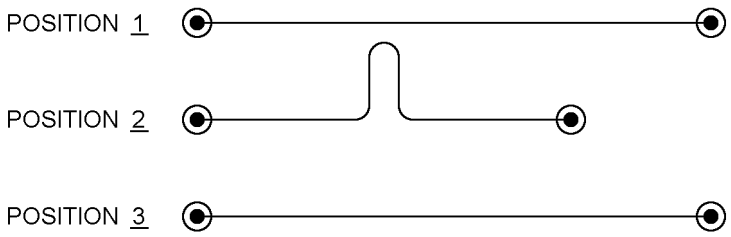
Lightning current: Not applicable to types A, B, C, D, E and F.

Dimensions and configuration: See figure 2.

Flexibility: For type “C” aluminum assemblies, change the MIL-DTL-83413 test to the following: Twist and flex: prior to the tensile strength test of MIL-DTL-83413.

Test No. 1 – Twist test 3 full twists in a direction to tighten the wire weave.

Test No. 2 – Flex 20 times.



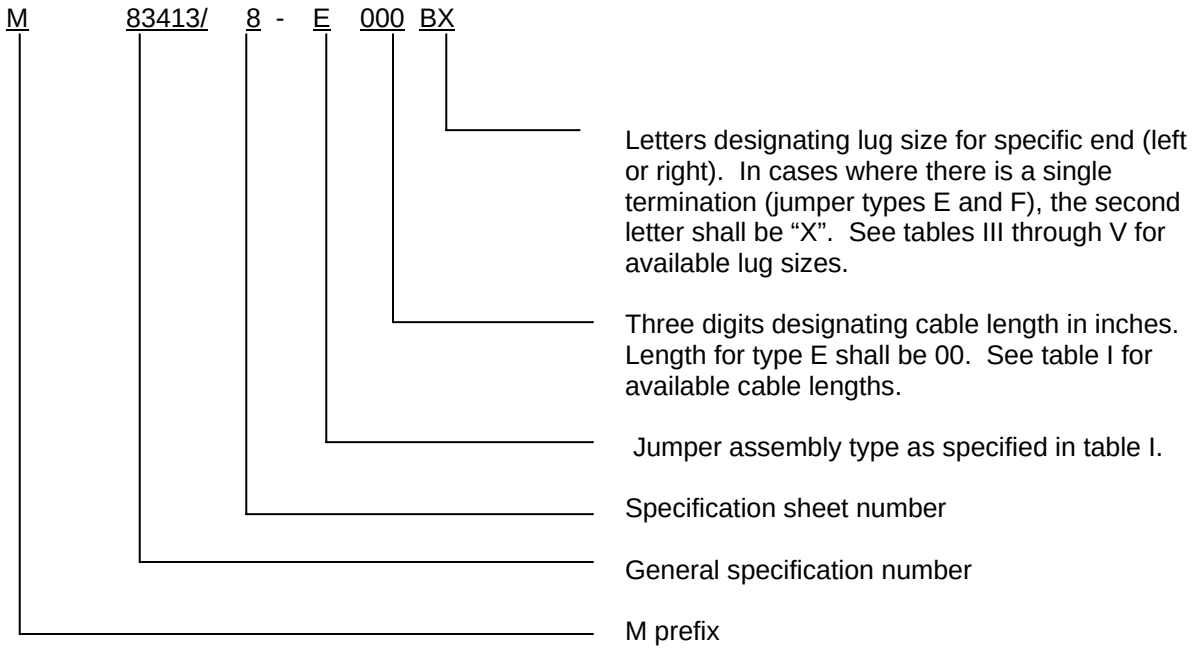
NOTE: For copper assemblies the flexibility test in accordance with MIL-DTL-83413 shall be performed at a rate of 30- 65 cycles/minute.

FIGURE 2. Twist and flex test.

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Qualification required: The activity responsible for the qualified products list for this specification sheet is the Defense Supply Center, Columbus, DSCC-VQ, P.O. Box 3990, Columbus, Ohio 43218-3990.

Example of PIN: M83413/8-A003AE



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Referenced documents. In addition to MIL-DTL-83413, this document references the following:

MIL-DTL-5541	SAE-AMS-QQ-A-225
MS23002	SAE-AMS-QQ-A-250
MS25441	SAE-AS25036
MS90485	SAE-AS20659
ASTM B230/B230M	SAE-AMS-DTL-23053/5
ASTM-B211	AWS C3.7
ASTM-B209	A-A-59551
EIA 364-26	
SAE-AS7928	
SAE-AS6852	

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CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 85
DLA-CC

Preparing activity:
DLA-CC

(Project 6150-2007-008)

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force - 02, 71, 99
DLA -GS

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